

SkillCraft

Local work, with a reliable record.

Employers post jobs. Artisans bid. The database keeps every decision, agreement and review connected.

WORKING PROTOTYPE

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01 A job is posted

02 An artisan is selected

03 A contract records the outcome

The problem is missing work history

Small jobs often live in calls and chat messages; decisions are hard to trace later.

Today: scattered messages

Who applied? Which price was accepted? Was payment acknowledged? The answers can be buried or disputed.

SkillCraft: linked records

The original job, every bid, the chosen artisan, the contract and the later review stay connected.

Who it serves

Local artisans seeking visible opportunities, and employers comparing offers for short projects.

DOMAIN: LOCAL EMPLOYMENT

SDG 8 alignment: access to decent work and a portable work record. Economic impact is a goal, not a measured result yet.

One job becomes one traceable agreement

The same workflow is visible from two different accounts.

1 POST

Employer adds a brief, location and budget.

2 DISCOVER

Artisan searches by skill and place.

3 BID

Artisan offers a price and optional note.

4 SELECT

Employer accepts one bid; other bids close.

5 ACKNOWLEDGE

Employer records payment sent; artisan confirms receipt.

6 REVIEW

Paid contract receives a review and updates trust score.

Each account sees the work relevant to it

The roles share the same underlying job and contract, but their menus and actions differ.

EMPLOYER / RAMESH

- Hiring overview
- Post a job and inspect bids
- Accept or decline an offer
- Record payment sent
- Review a paid contract

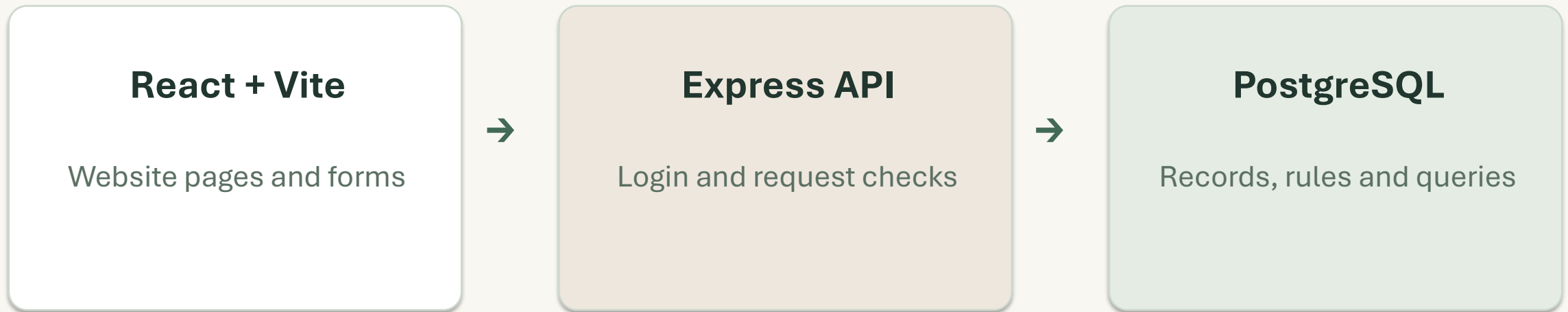
ARTISAN / LAKSHMI

- My workspace
- Find work and submit a bid
- Track bid status
- Confirm payment received
- See agreements and reviews

The API also checks the role and ownership; the database applies row-level rules for private bids, contracts and events.

The database does the critical work

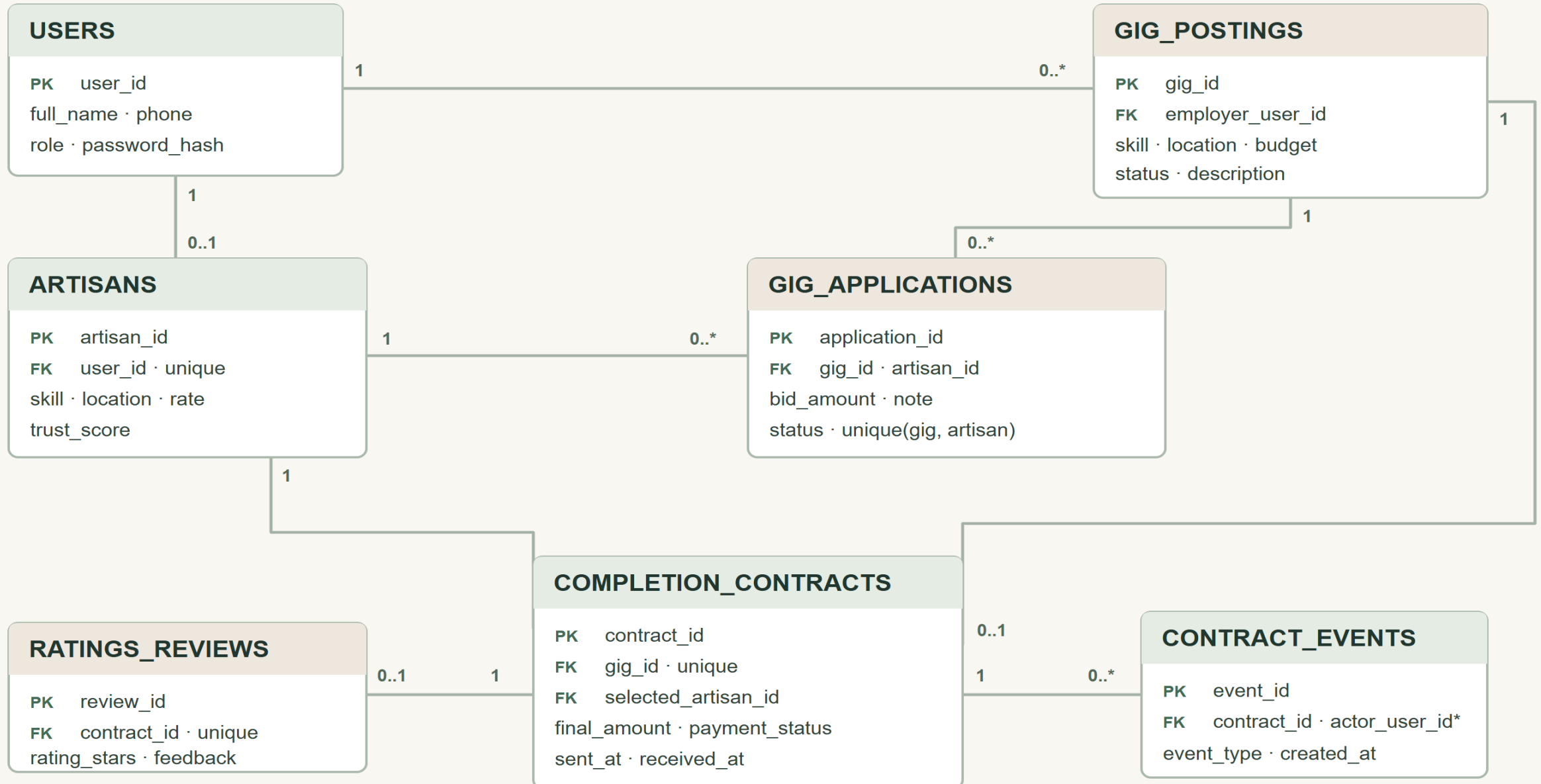
A plain-language map of the technology stack.



Local preview: PGLite runs the PostgreSQL-compatible schema. Docker configuration uses PostgreSQL 15.

SkillCraft · relational model

PK primary key · FK foreign key · 1 / 0..* cardinality



* actor_user_id references USERS and can be null for legacy events. Accepting a bid creates a contract through a trigger, without a direct bid-to-contract FK.

Rules protect the data when people act at once

These protections are in SQL and transactions, not only in button logic.

Keys + checks

One bid per artisan per job; one contract per job; positive amounts; valid statuses.

Safe acceptance

A transaction locks the job row. Accepting one bid closes the job and rejects competitors.

Triggers + functions

Contract creation, event history and trust-score refresh happen in the database.

Two confirmations

A new contract cannot be marked paid until payment-sent and receipt timestamps both exist.

The innovation is a verifiable sequence of claims

Conventional status flags say 'paid'; SkillCraft records who said what and when.

CONVENTIONAL APPROACH

**Employer flips one
payment flag to paid.**

The worker's acknowledgment is missing.

SKILLCRAFT NOW

**Employer records sent.
Artisan confirms received.**

An append-only history records each event.

A review is allowed only after a new contract is paid. Older paid rows remain visibly legacy, not falsely backfilled.

A review updates a careful trust score

Only a paid contract can receive a review. A trigger refreshes the artisan's stored score.

ONE NEW 5-STAR REVIEW

$$(5 + 3 \times 3.5) \div (1 + 3) = 3.88$$

ELIGIBLE DATA

Reviews tied to paid contracts.

STABILIZING PRIOR

Three virtual 3.5-star reviews.

AUTOMATIC UPDATE

Review insert runs the score function.

Matching and privacy are database features

The discovery screens are backed by indexed queries and restricted database access.

SEARCH

Jobs and artisans can be filtered by skill and searched by place or keyword. B-tree and full-text indexes support those queries.

EXPLAINABLE FIT

A simple 0–3 score prefers the same skill (2 points) and same city (1 point). It is inspectable, not an AI prediction.

ROW SECURITY

A restricted database role and row-level policies limit access to a person's private bids, contracts and events.

Public artisan reputation appears through limited views; private account fields stay private.

What we have actually validated

Functional evidence is strong; performance and social impact still need measurement.

API + SCHEMA

Checks the full job-to-review flow, foreign-key relationships, audit immutability, and ownership rules.

24 JEST CASES

20 CRUD cases plus 4 concurrency and integrity cases passed on fresh local PGlite databases.

2 BROWSER FLOWS

The two-role payment journey and long-text mobile layout passed in Edge.

Next evidence: benchmark indexed search vs. a baseline on native PostgreSQL; measure fit quality with labeled examples.

What the panel grades • part 1

The numbers are available marks, not a score awarded to this project.

4**Problem + domain**

Informal local hiring lacks a reliable, portable record.

5**Database design**

Seven linked tables, ER diagram, keys, constraints and normalization.

5**DBMS depth**

Transactions, triggers, functions, indexes, views and row-level security.

4**Innovation**

Two-party acknowledgment plus append-only contract events.

What the panel grades • part 2

The claim and the evidence should stay at the same level.

5**Novelty**

Compare with employer-only payment and a plain rating average.

2**SDG impact**

SDG 8 goal: fairer access and a portable history; impact unmeasured.

3**Validation**

Show passing tests; be honest that benchmark data is pending.

2**TRL + demo**

Run the two-account prototype live; no field pilot yet.

A five-minute live demonstration

Use the connected local preview, not the browser-only demo mode.

- | | | |
|---|--------------------------------|--------------------------------------|
| 1 | Ramesh posts | ramesh@constructionco.in |
| 2 | Lakshmi searches + bids | lakshmi@skillcraft.local |
| 3 | Ramesh accepts | One job closes; one contract appears |
| 4 | Both acknowledge | Sent → received → paid |
| 5 | Ramesh reviews | Show event history and trust score |

BOTH PASSWORDS: password123

A working prototype with clear next measurements

The database is the product's backbone; our claims stop where the evidence stops.

WHAT WORKS NOW

- Connected two-role website
- Seven-table relational model
- One-winner bid acceptance
- Two-party payment record
- Contract-linked trust and audit history

WHAT WE WILL MEASURE NEXT

- Native PostgreSQL query latency
- Matching quality on labeled jobs
- Security checks on a native server
- User outcomes in a real pilot

SkillCraft records payment claims. It does not send money, verify banks, or prove an earnings uplift.